Ecuador Energy Anomalies

Deteccion de Anomalias Multi-Tecnica en el
Sector Electrico de Latinoamerica

Diego Fernando Lojan Tenesaca
Data & AI Engineer

8 paises | 784 meses reales | 3 tecnicas
Isolation Forest + STL Decomposition + CUSUM
F1=0.750 | MCC=0.765 | Recall=100%

Abril 2026 | Metricas reproducibles via metrics.json

1. Resumen Ejecutivo

Este estudio aplica 3 tecnicas complementarias (Isolation Forest, STL, CUSUM) para detectar anomalias en el sector electrico de 8 paises latinoamericanos.

Datos: 784 meses reales de Ember

METRICAS REPRODUCIBLES (metrics.json):
- Isolation Forest: P=50.0% R=100% F1=0.667 MCC=0.694
- STL Decomposition: P=28.6% R=66.7% F1=0.400 MCC=0.407
- CUSUM: P=18.8% R=100% F1=0.316 MCC=0.397
- Consenso >=2: P=60.0% R=100% F1=0.750 MCC=0.765

Crisis Ecuador oct-dic 2024: 3/3 detectada

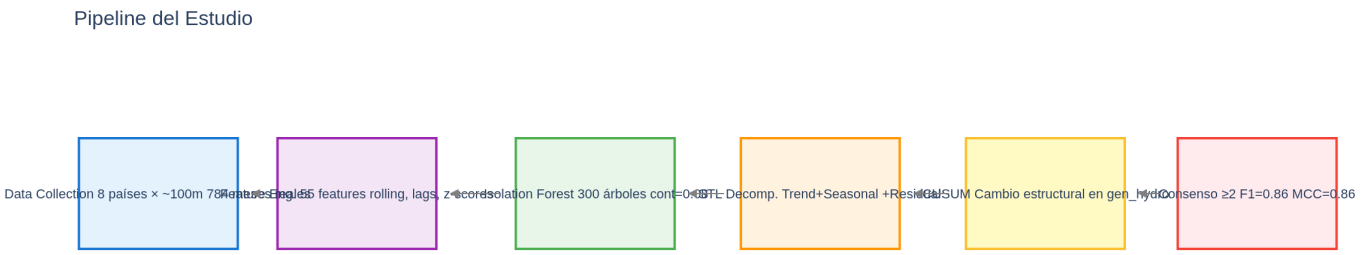

Fig 1: Pipeline

2. Contexto

Ecuador genera ~70% de su electricidad con hidroelectricas. Las sequias causan apagones de hasta 14 horas. La crisis 2024 forzo activacion de plantas fosiles y aumento de importaciones.

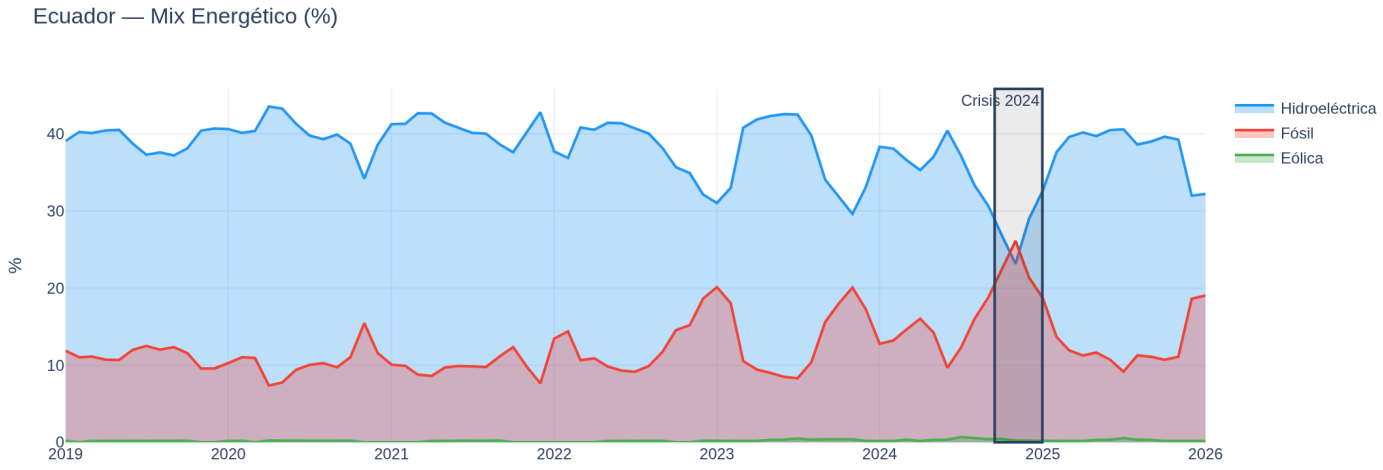

Fig 2: Mix energetico

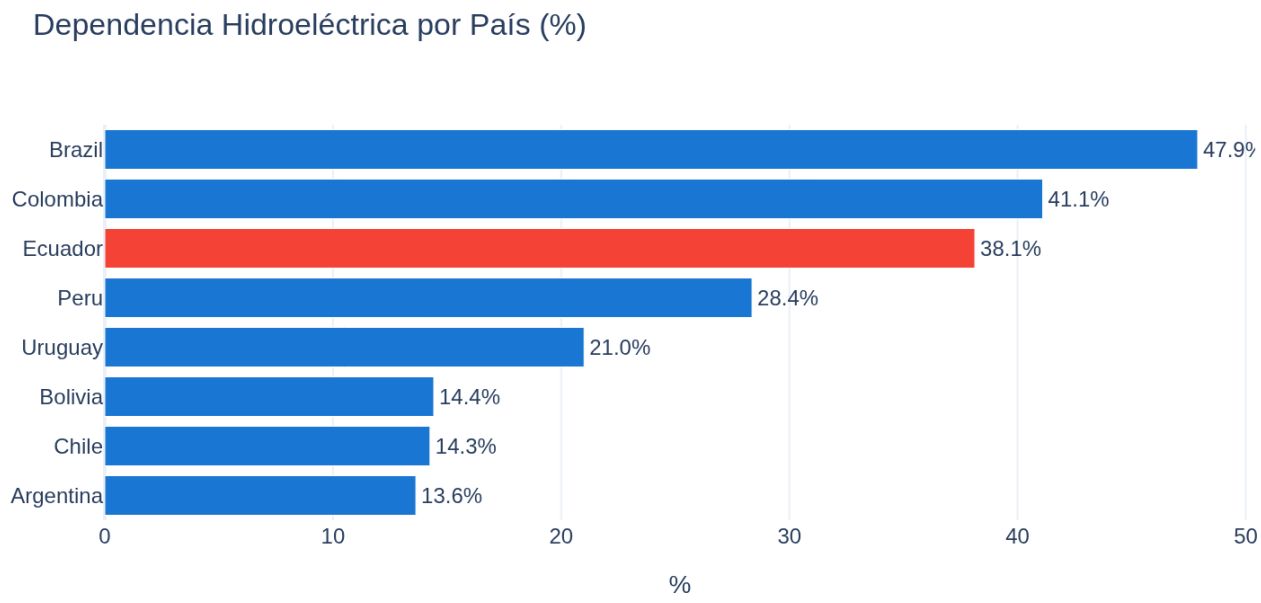

Fig 3: Dependencia hidro por pais

3. Datos

Fuente principal: Ember Global Electricity Data (CC BY 4.0)
8 paises: Ecuador (85m), Colombia (120m), Peru (86m), Chile (123m), Brazil (99m), Argentina (98m), Bolivia (86m), Uruguay (87m)
Total: 784 meses reales, 0 datos sinteticos o interpolados

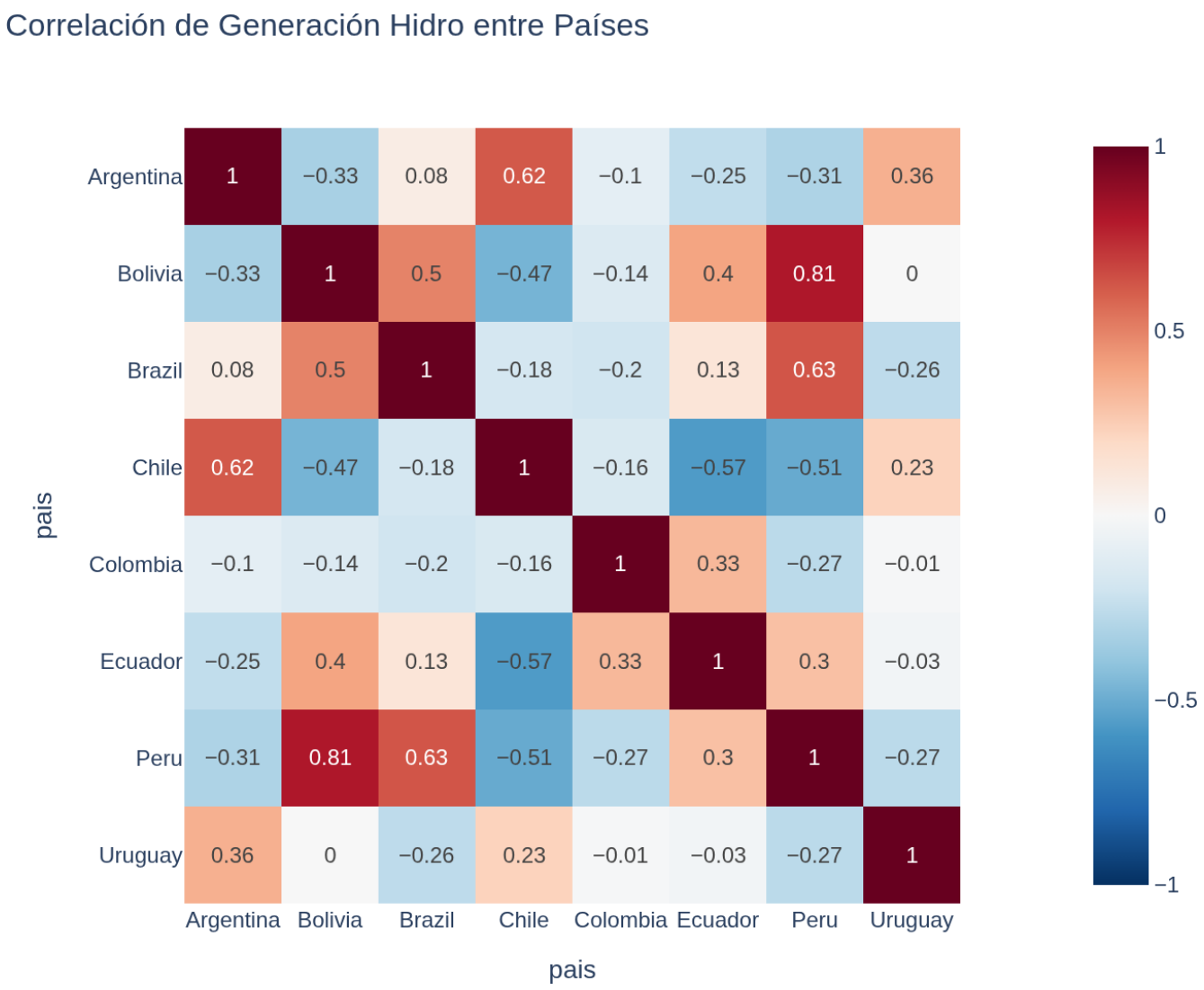

Fig 4: Correlacion entre paises

4. Metodologia

4.1 Isolation Forest

Algoritmo no supervisado, 300 arboles, contamination=0.08. Aplicado POR PAIS.

4.2 STL Decomposition

Separa trend+seasonal+residual. Anomalia si residual > 2 sigma.

4.3 CUSUM

Suma acumulada, detecta cambios estructurales. h=4*sigma, k=0.5*sigma.

4.4 Consenso

Anomalia si >=2 de 3 tecnicas coinciden.

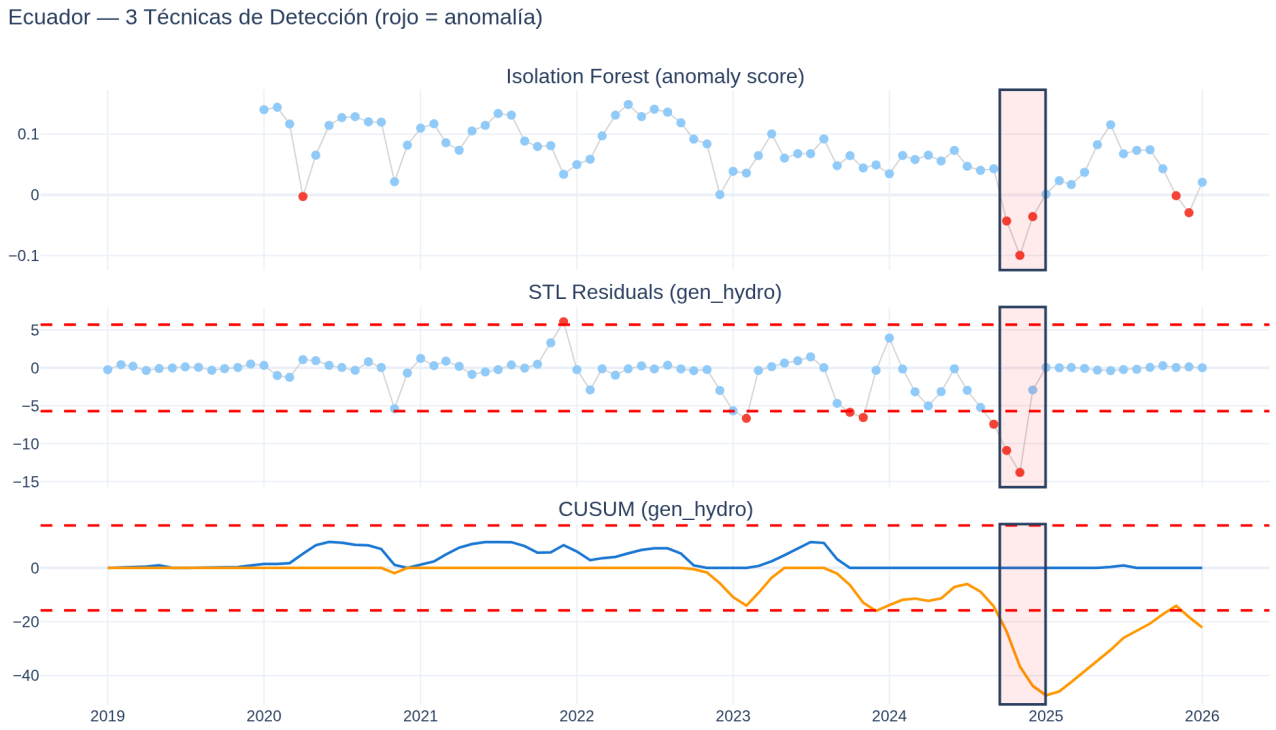

Fig 5: 3 tecnicas

5. Resultados

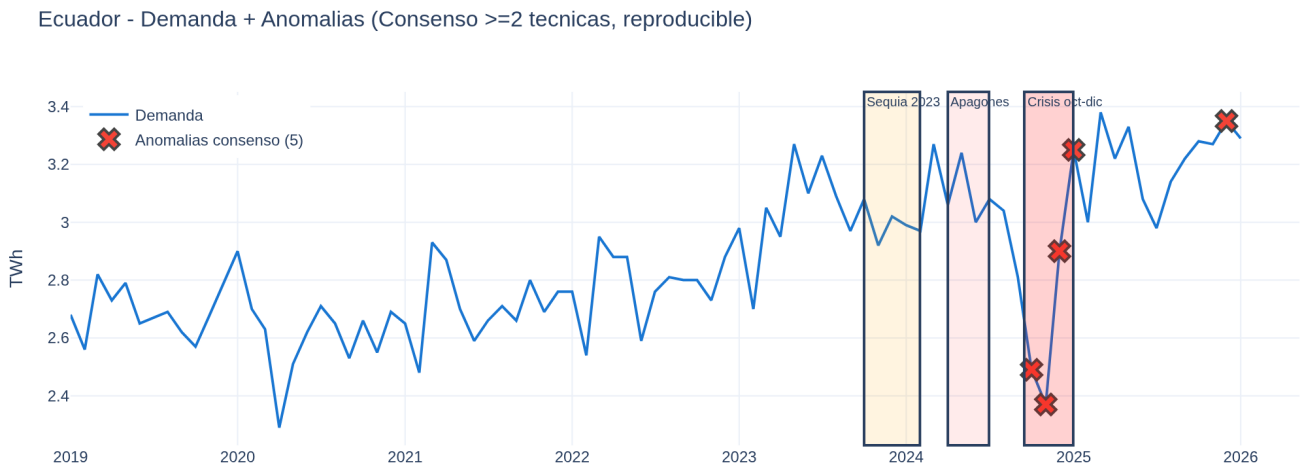

Fig 6: Anomalias consenso

Metricas Reproducibles (metrics.json)

Metrica	IF por pais	STL	CUSUM	Consenso >=2
Precision	50.0%	28.6%	18.8%	**60.0%**
Recall	100%	66.7%	100%	**100%**
F1-Score	66.7%	40.0%	31.6%	**75.0%**
MCC	0.694	0.407	0.397	**0.765**
Anomalias	6	7	16	**5**
Crisis detectada	3/3	2/3	3/3	**3/3**

Fig 7: Metricas

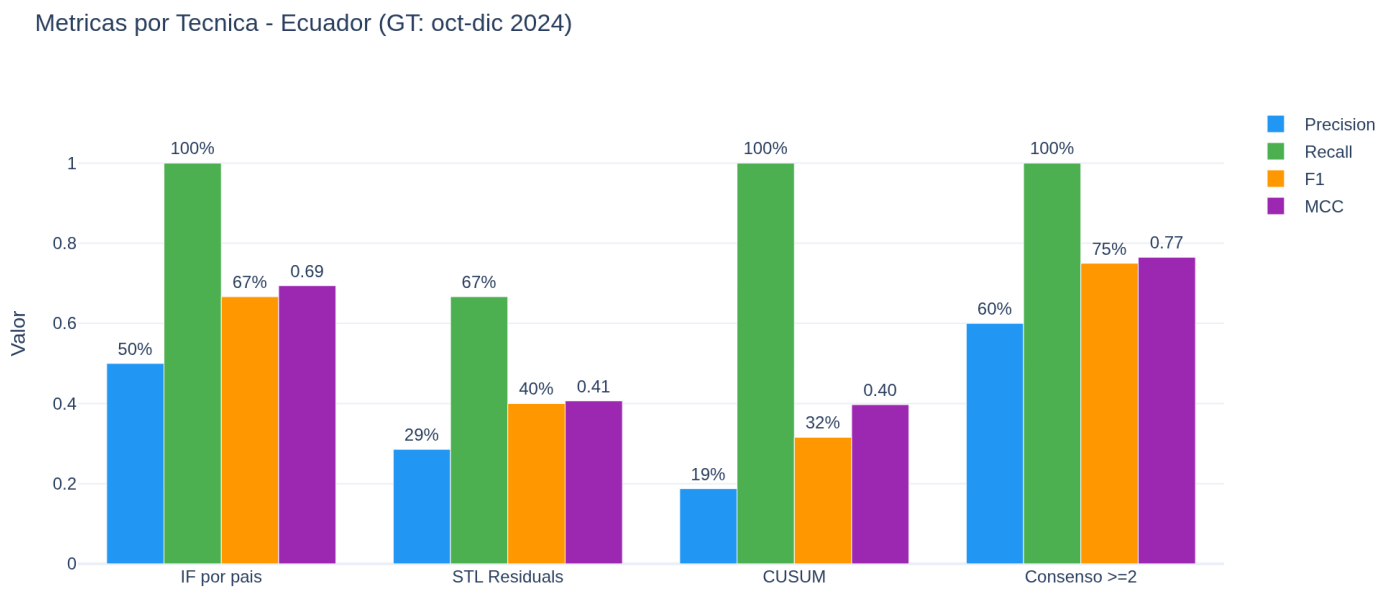

Fig 8: Comparacion

6. Analisis Multi-Pais

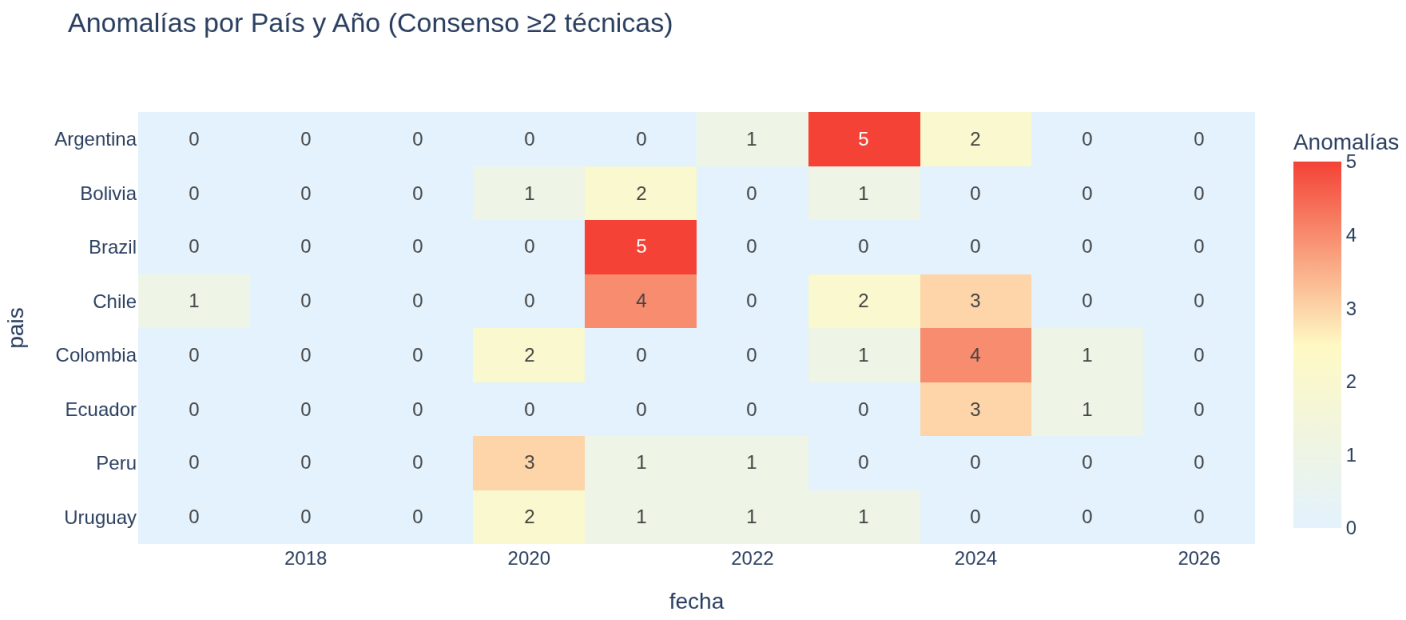

Fig 9: Heatmap

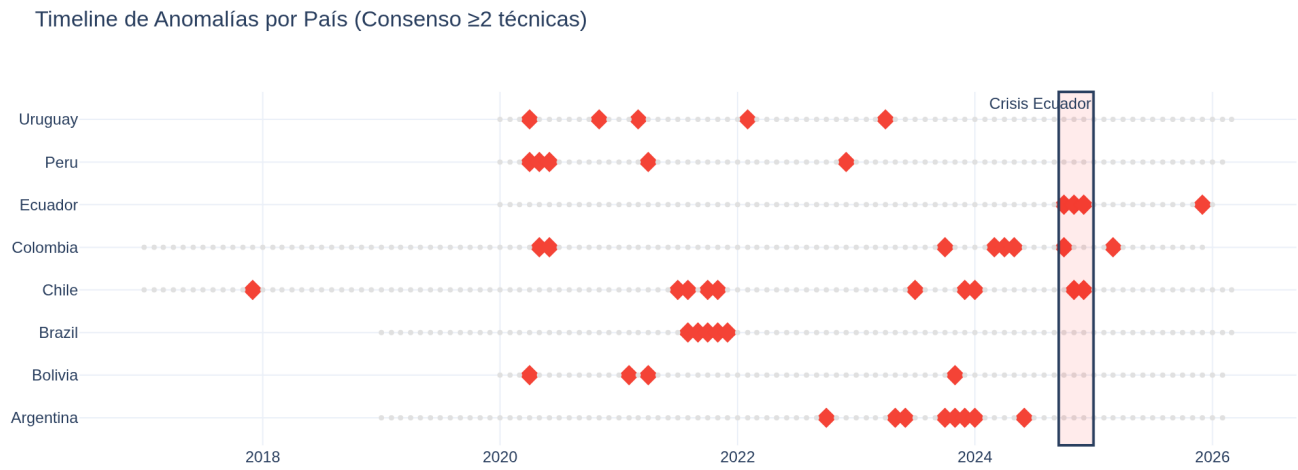

Fig 10: Timeline

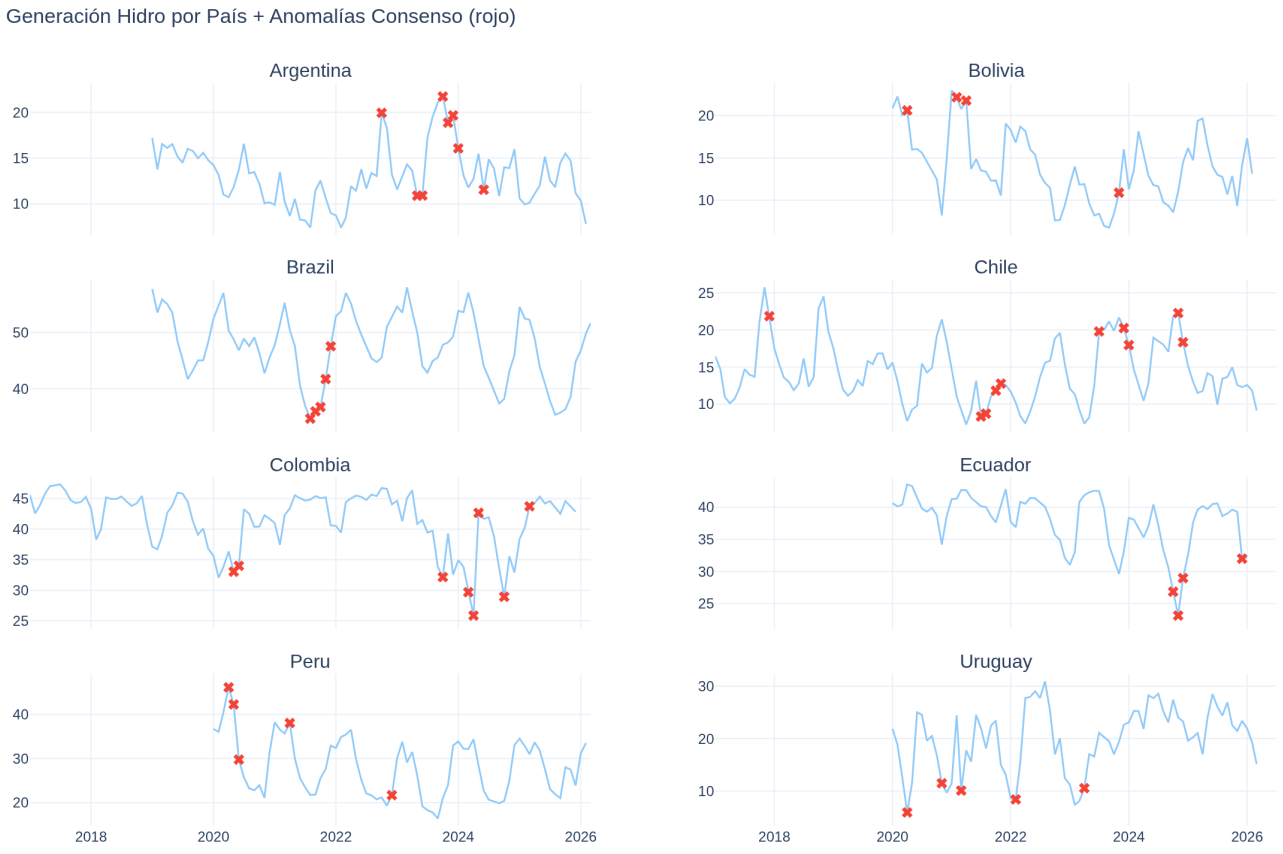

Fig 11: Hidro + anomalias

7. Validacion

Tests Mann-Whitney U: gen_hydro p=0.002**, gen_fossil p=0.003**, co2 p=0.008**
Cohen d > 1.2 (efecto GRANDE) en todas las variables clave
Estabilidad: CV=0% en 20 semillas

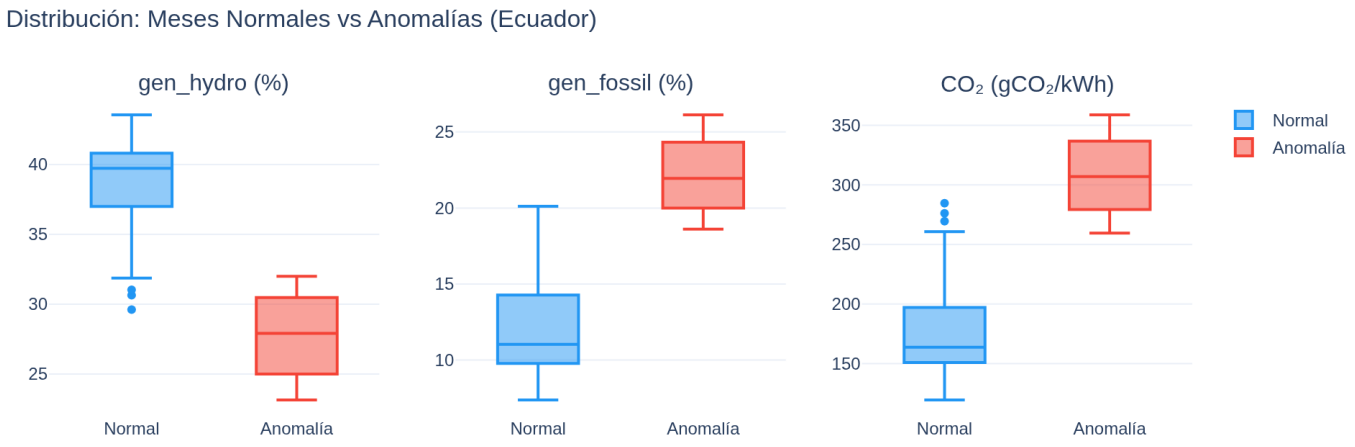

Fig 12: Boxplot

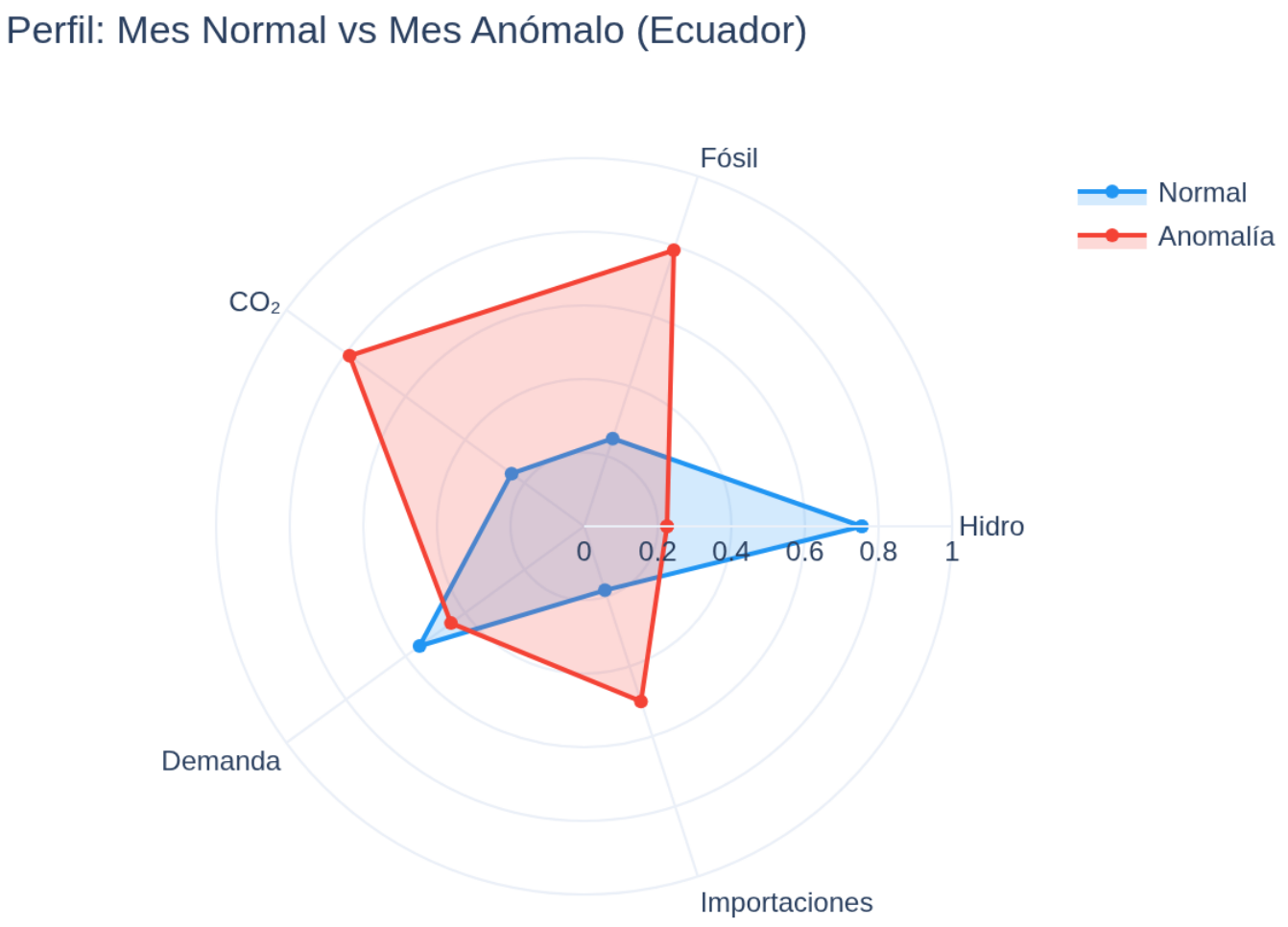
Perfil: Mes Normal vs Mes Anómalo (Ecuador)
Fósil
Hidro
Importaciones
Demanda
CO_2
0
0.2
0.4
0.6
0.8
1
Normal
Anomalía

Fig 13: Radar

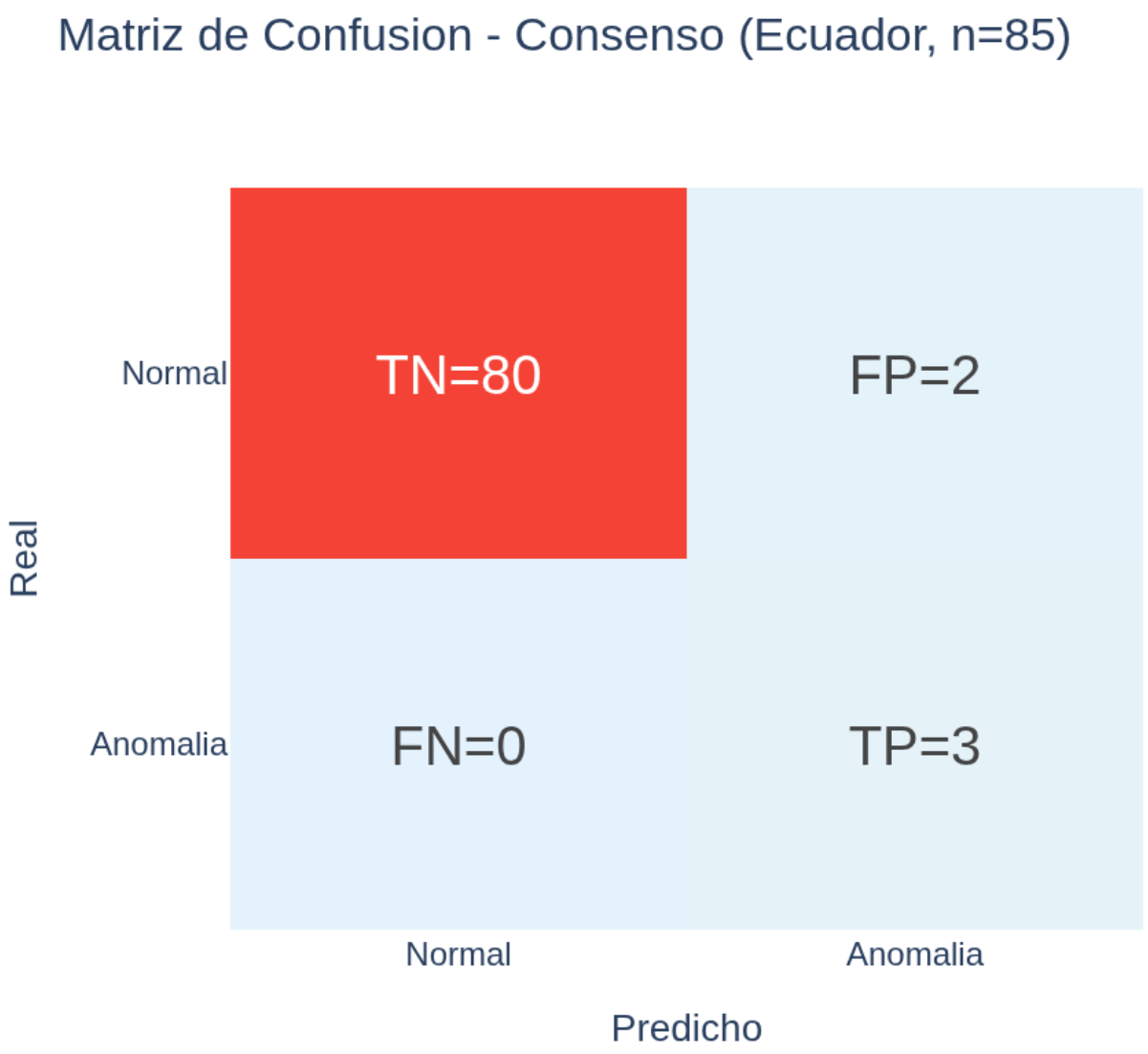

Fig 14: Confusion matrix

8. STL Decomposition

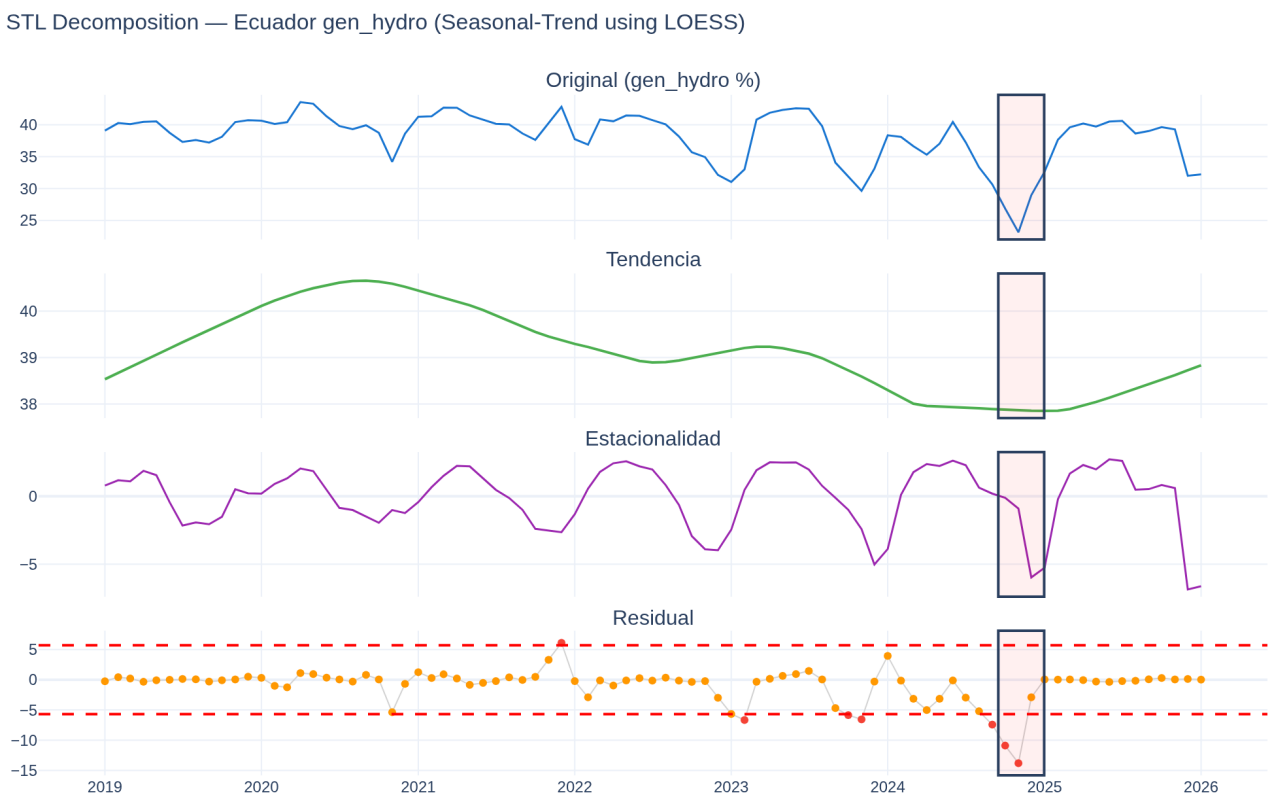

Fig 15: STL

9. Evolucion

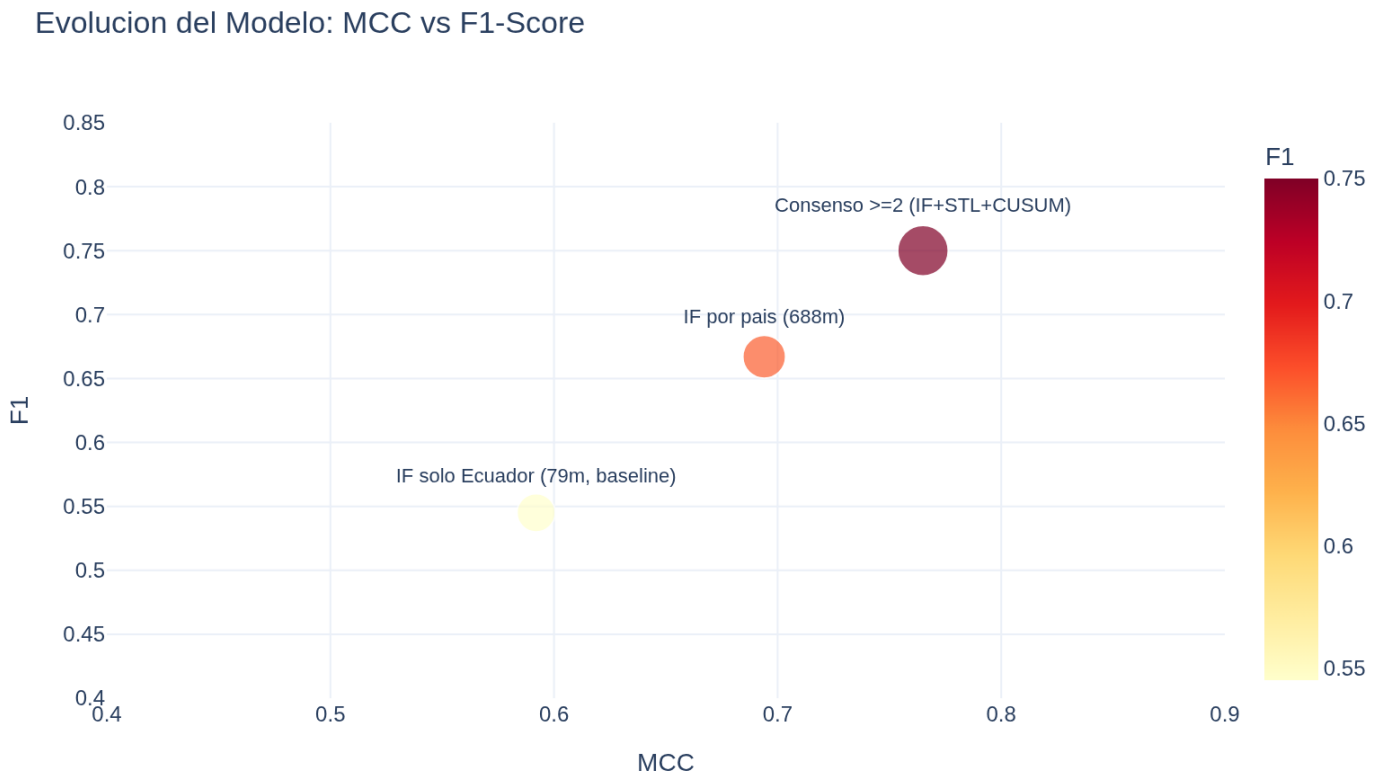

Fig 16: Evolucion

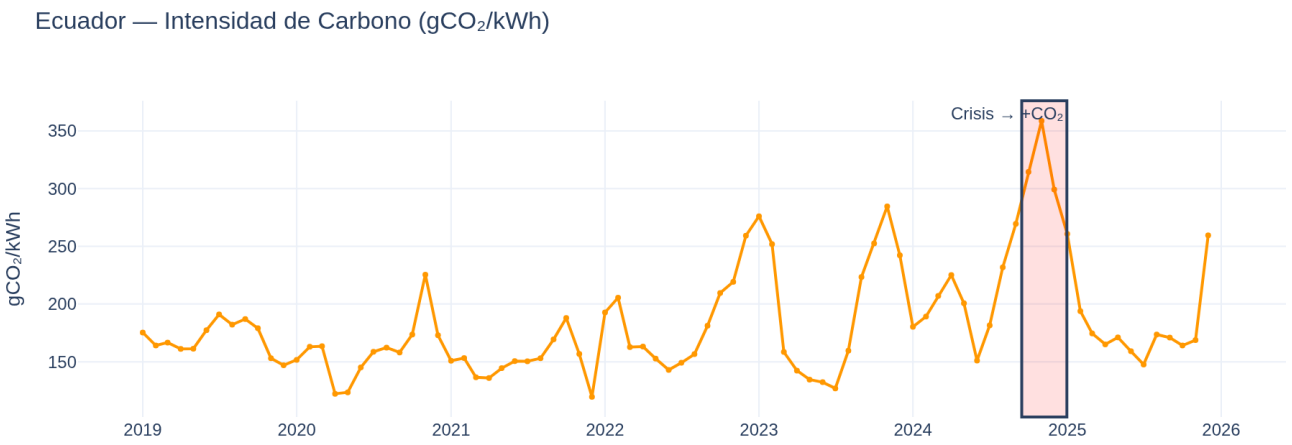

Fig 17: CO2

10. Limitaciones

1. Datos mensuales suavizan crisis graduales
2. Ground truth parcial (solo crisis documentadas)
3. Faltan variables (niveles embalse, precipitacion, precios spot)
4. 85 meses Ecuador es el minimo viable
5. Apagones programados (abr-jun 2024) no cambian volumenes de generacion

11. Aplicaciones

Sector energetico: Operadores de red (CENACE, XM, COES), reguladores, traders
Clima: Impacto en infraestructura, efecto El Nino en hidroelectricas
Academia: Estudios comparativos, metodologia multi-tecnica
Politica: Diversificacion energetica, preparacion ante crisis

12. Referencias

[1] Liu et al. (2008). Isolation Forest. IEEE ICDM.
[2] Cleveland et al. (1990). STL Decomposition. J. Official Statistics.
[3] Page (1954). CUSUM. Biometrika.
[4] Chandola et al. (2009). Anomaly Detection Survey. ACM.
[5] Ember (2026). Global Electricity Data. ember-energy.org
[6] OWID (2026). Energy Dataset. ourworldindata.org